

RIMAC GYM

AI of Byte

Context & Field Site

Field Site

- UCSD RIMAC Treadmill Area
- Peak Hours:
12:00 - 14:00
18:00 - 20:00

Users

- **Active users:** students currently using treadmills
- **Waiting users:** students standing near specific machines
- **Scanning users:** students looking across the space for availability



Design Problem

Student Must Make Decisions Without Clear Information

Current Situation

- A 20-minute usage rule exists when others are waiting
- The rule is rarely followed or clearly enforced
- There is no visible queue or availability system

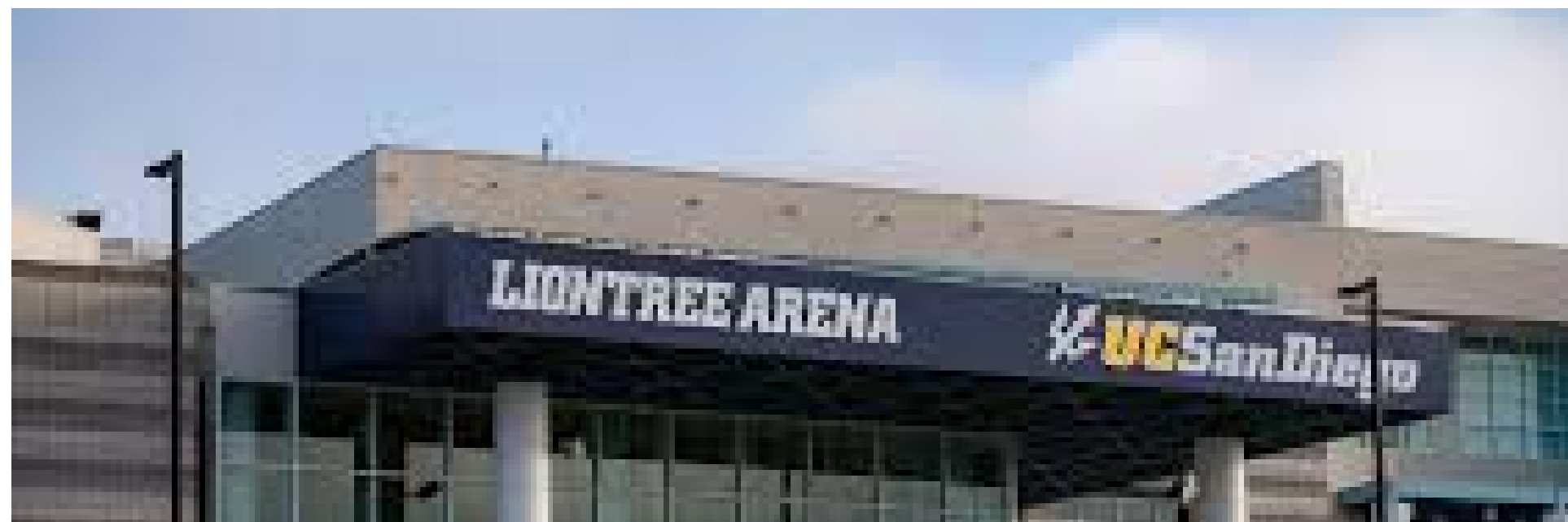
Users Lack Information About

- Equipment availability
- Waiting conditions
- Who is next

As a result

Impact

- Uncertainty
- Social awkwardness
- Inefficient waiting
- Inconsistent access to equipment



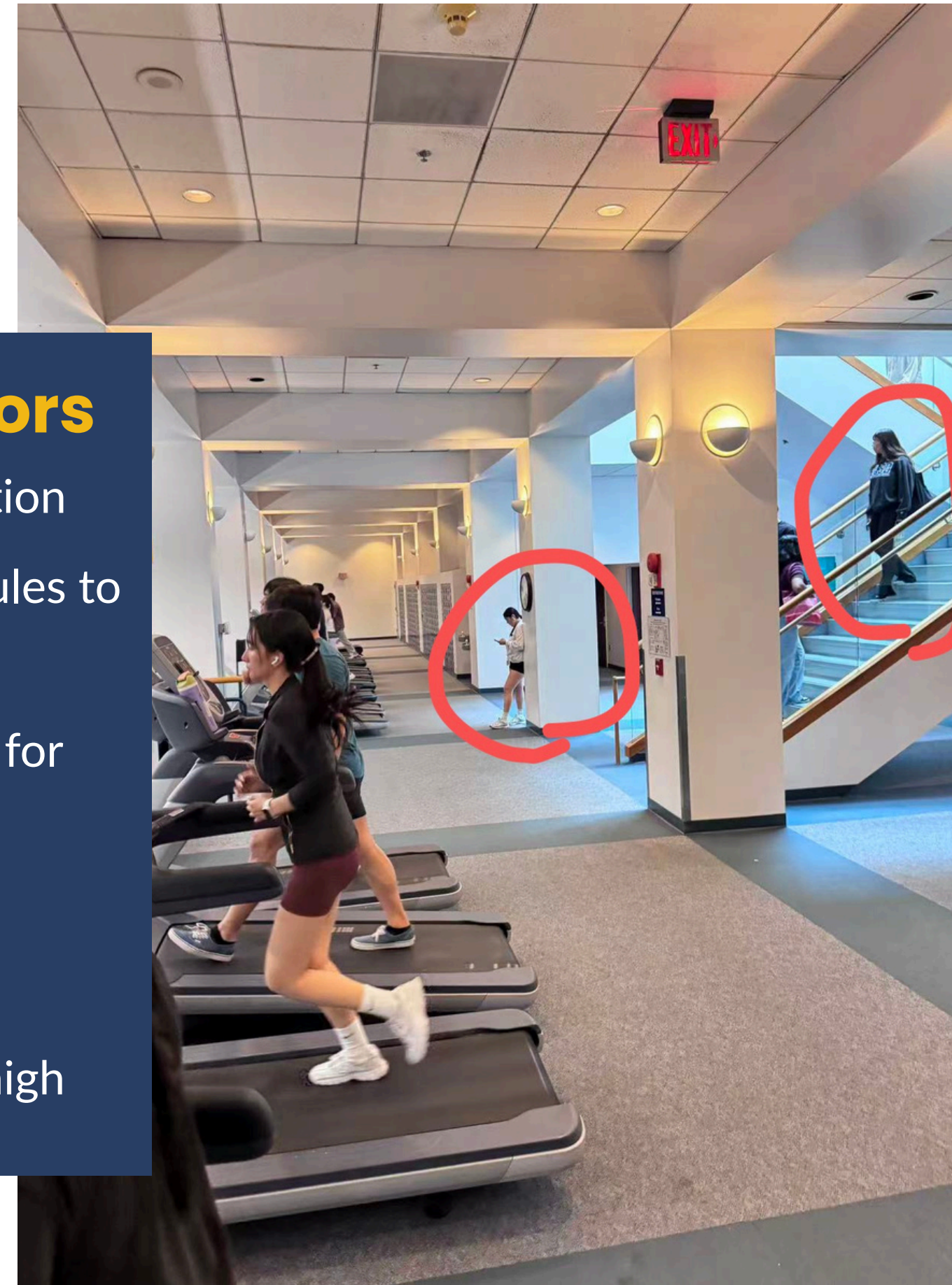
What We Observed

Current Waiting Experience

- 🕒 Peak-hour congestion
- 🚶 Students repeatedly walk around checking machines
- ❌ No visible queue or waiting system
- 🚶 Waiting users spread throughout the space
- 🙋 Availability is determined through observation and guesswork

Common Behaviors

- 😬 Avoiding direct interaction
- 📝✎ Change workout schedules to avoid crowds
- 🔍 Walk around searching for available treadmills
- ? Guessing who is next
- 🙌 Give up waiting when uncertainty becomes too high



User Interview Findings

What Actual User Told Us

Pain Points

- **Unclear** waiting order
- **Unpredictable** wait times
- Social discomfort when approaching others
- **Lack of reliable** availability information
- No visibility of the **20-minute rule**



“I often don’t know **how long I need to wait**. I never know **how long someone has been on the treadmill.**” — Vic

“Sometimes I just **give up and leave.**” — Jimmy



“I usually wait in open areas because **standing behind others feels stressful.**” — Mary



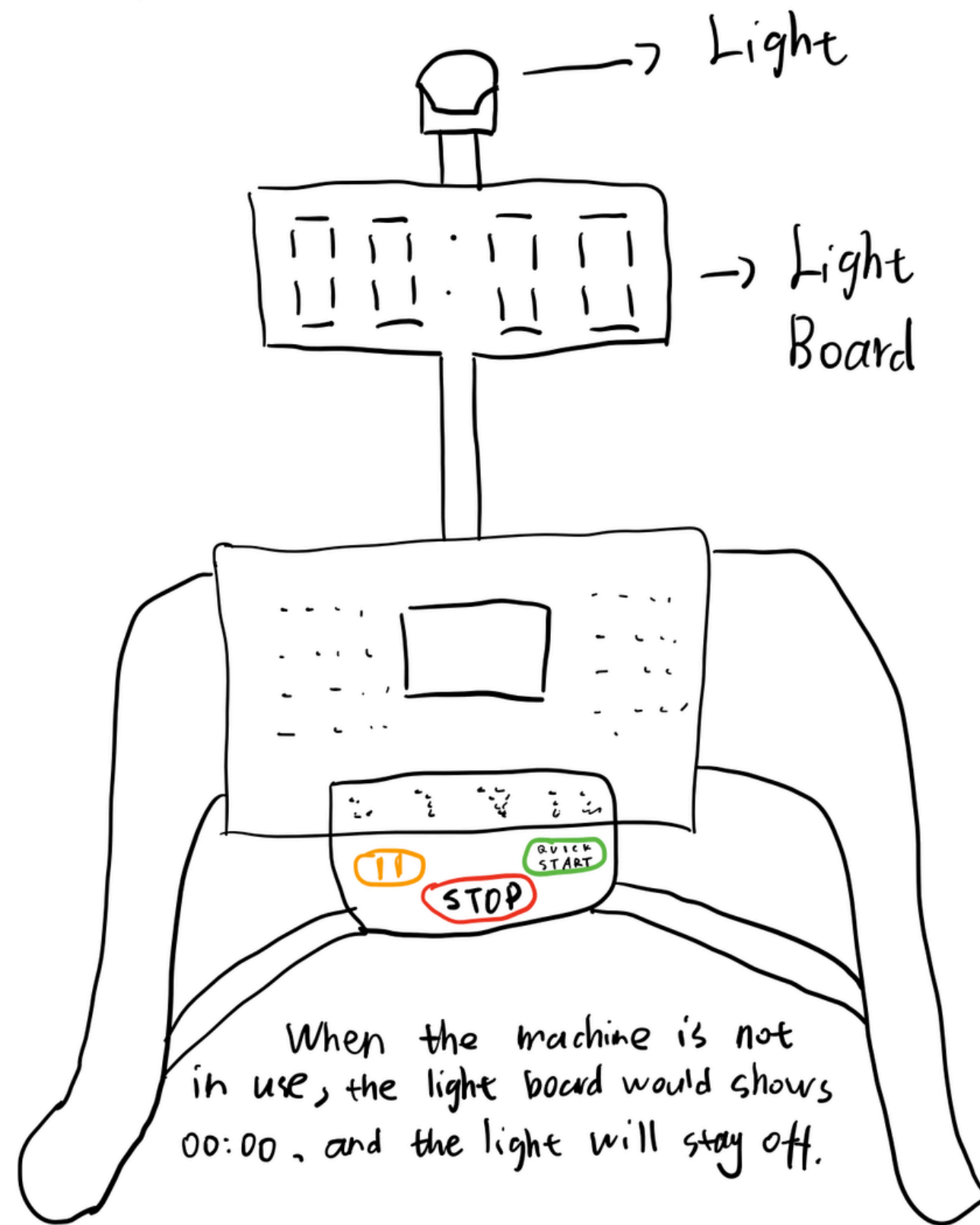
DESIGN **SOLUTION**

Solution-Button

Light indicator

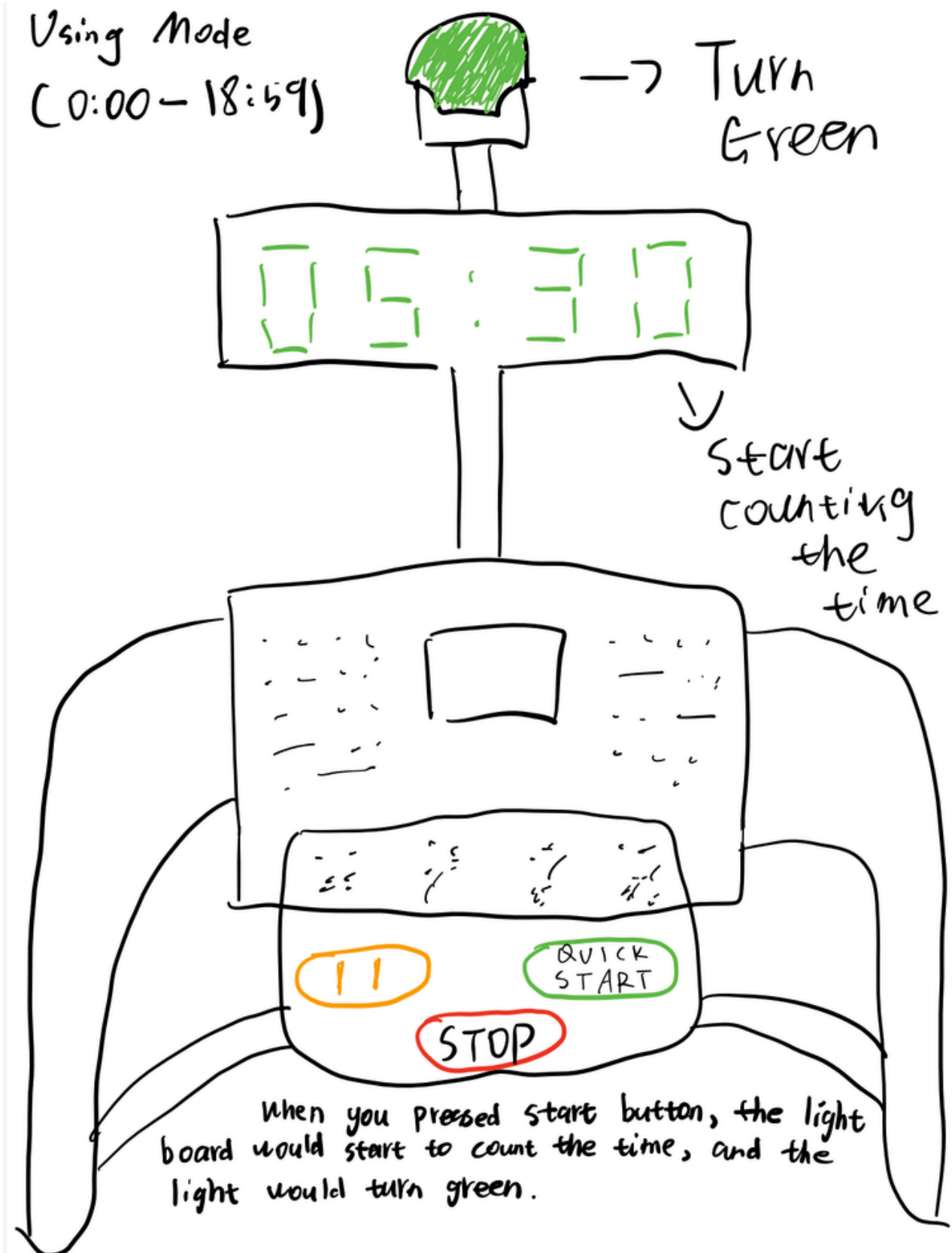
20min countdown timer

Available Mode



Using Mode

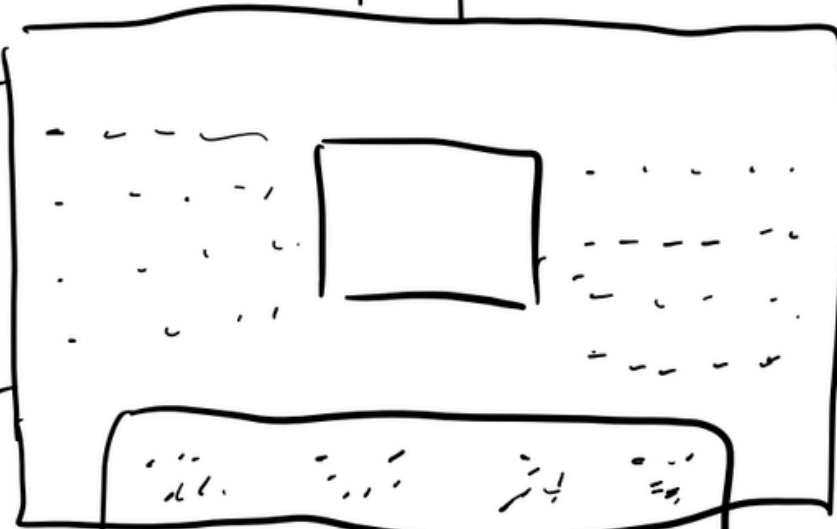
(0:00-18:59)



About To Overtime Mode
(19:00)



→ Turn red



Quick
START

STOP

When the time about to reach to the time limit, the light would turn red to notify the user.

Overtime Mode
(20:00 +)



→ Light
start
flashing



↘ Light
board
remain
counting

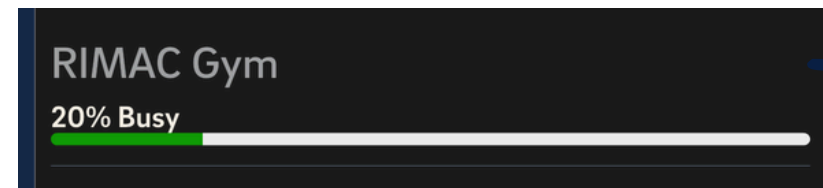


Quick
START

STOP

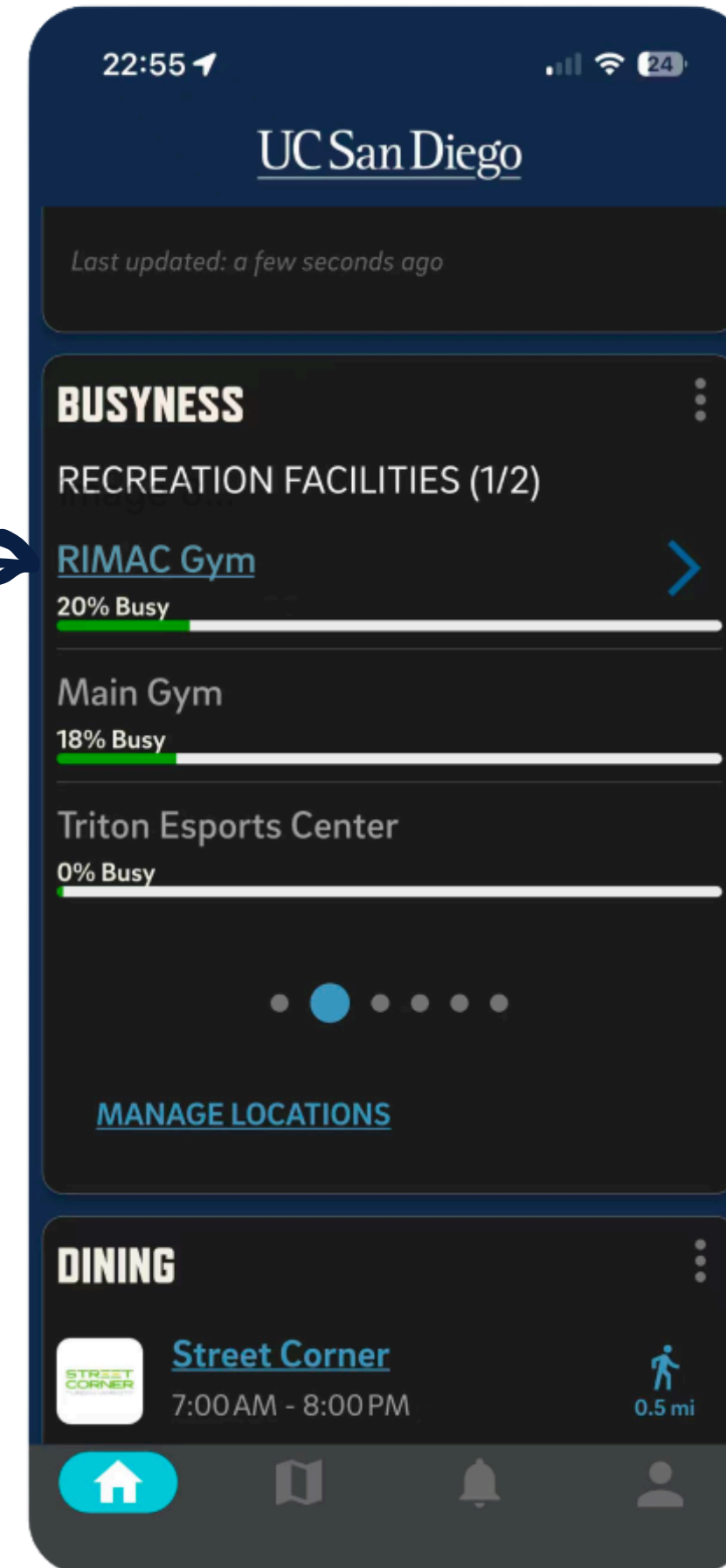
When the time reach to 20 min and get over it, the light board will keep counting the time, and the light would start flashing.

Solution 1 – UC SanDiego APP



Integrated into the UC San Diego Student App, this feature allows users to instantly check overall treadmill occupancy before heading to the gym.

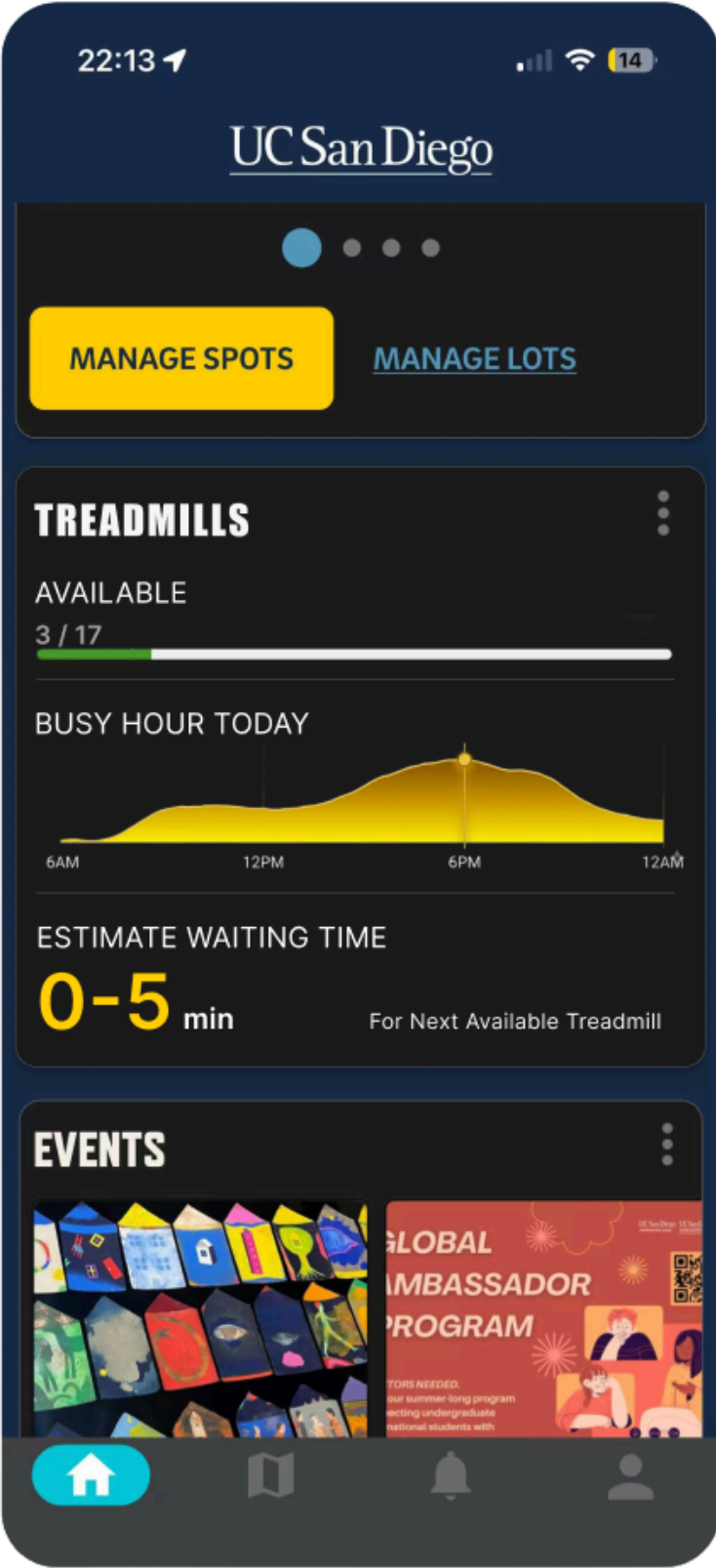
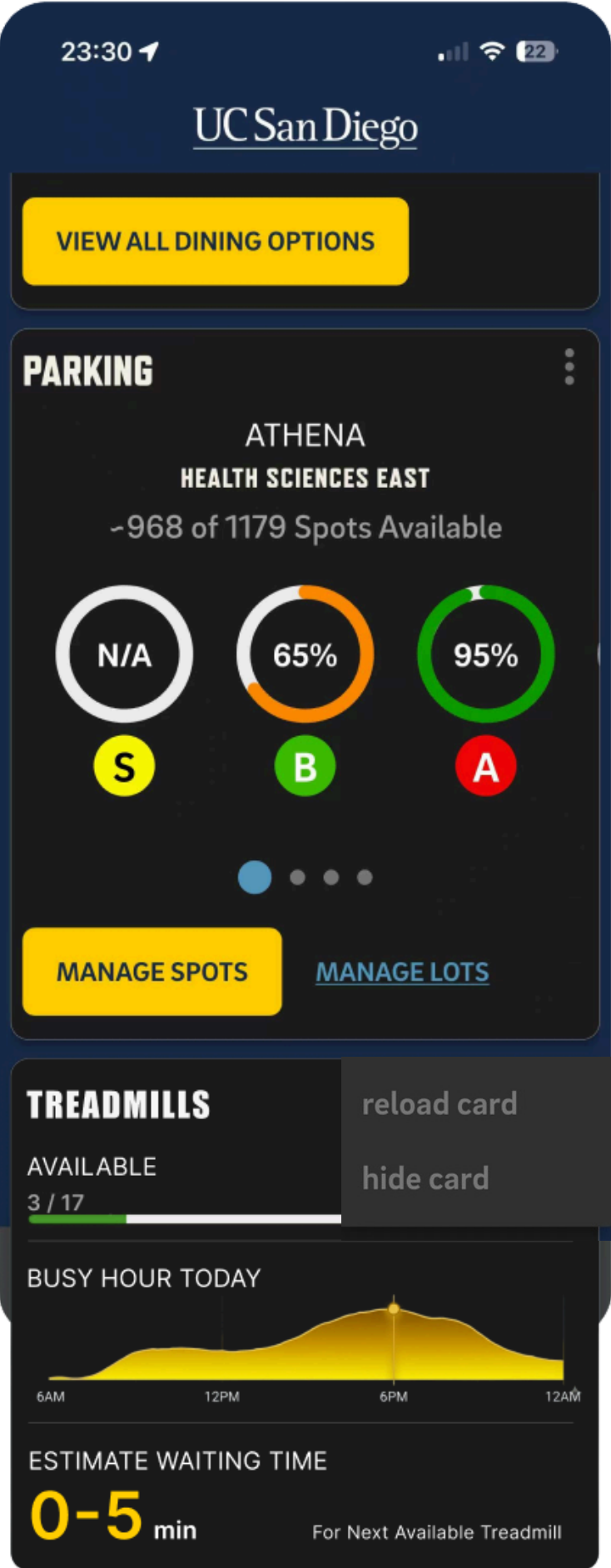
- Adaptation strategy



UC SanDiego APP

This helping users quickly assess availability before arriving at the gym.

- Pressure / Stress
- Waiting frustration
- Overcrowding



GROUP 15 - AI OF BYTE

Outlook

Design mockup displayed on a real smartphone screen.



Solution 2- TV Screen

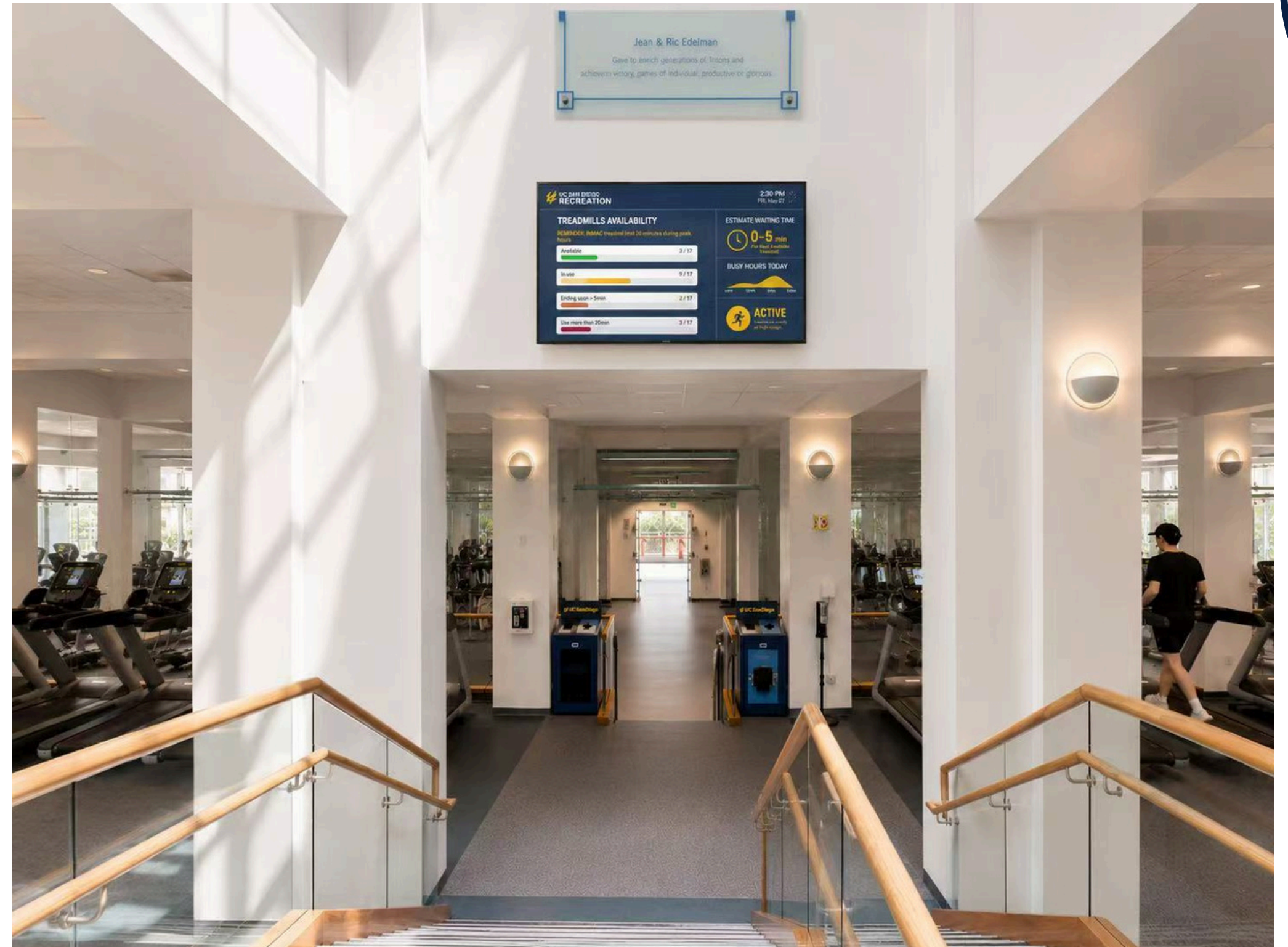
Located at the entrance of Rimac, this reducing uncertainty and improving the gym experience.

- Waiting behavior
- Social discomfort
- Waiting frustration
- Pressure / stress
- Space organization



Outlook

Mockup showing how the interface would appear on the TV display at the entrance of Rimac.



Design Solution – Experience Models

Before Arriving to RIMAC

- Check Availability in UCSD App 📱

Addresses

- Uncertainty about treadmill availability
- Arriving during busy periods

Breakdown

- Users can check treadmill availability before heading to the gym and better plan when to visit.

Arriving at RIMAC

- Real-Time Occupancy Display 📺

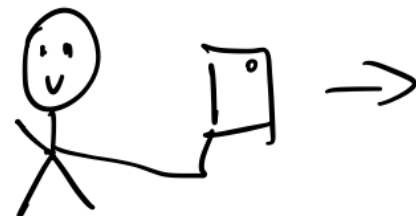
Addresses

- Searching for available treadmills
- Limited visibility of machine availability

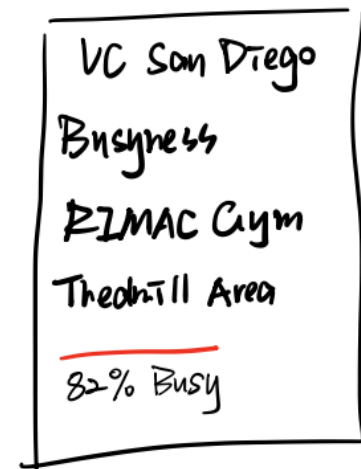
Breakdown

- The display provides immediate information about treadmill availability, reducing the need to walk around and search.

① Check on my phone



Before I go into RIMAC, I go and check the availability of the treadmill area on UCSD mobile app.



② Arrive at RIMAC.



Treadmill Availability		
Reminder: 20min rule.		
available		3/17
In use		5/17
ending soon		5/17
use more than 20 min		4/17

Design Solution – Experience Models

Waiting for a Treadmill

- Countdown Timer & Status Lights

Addresses

- Uncertainty about wait times
- Social discomfort while waiting

Breakdown

- Users can see how much time remains on each treadmill without needing to interrupt other gym users.

Using a Treadmill

- Occupancy Button System

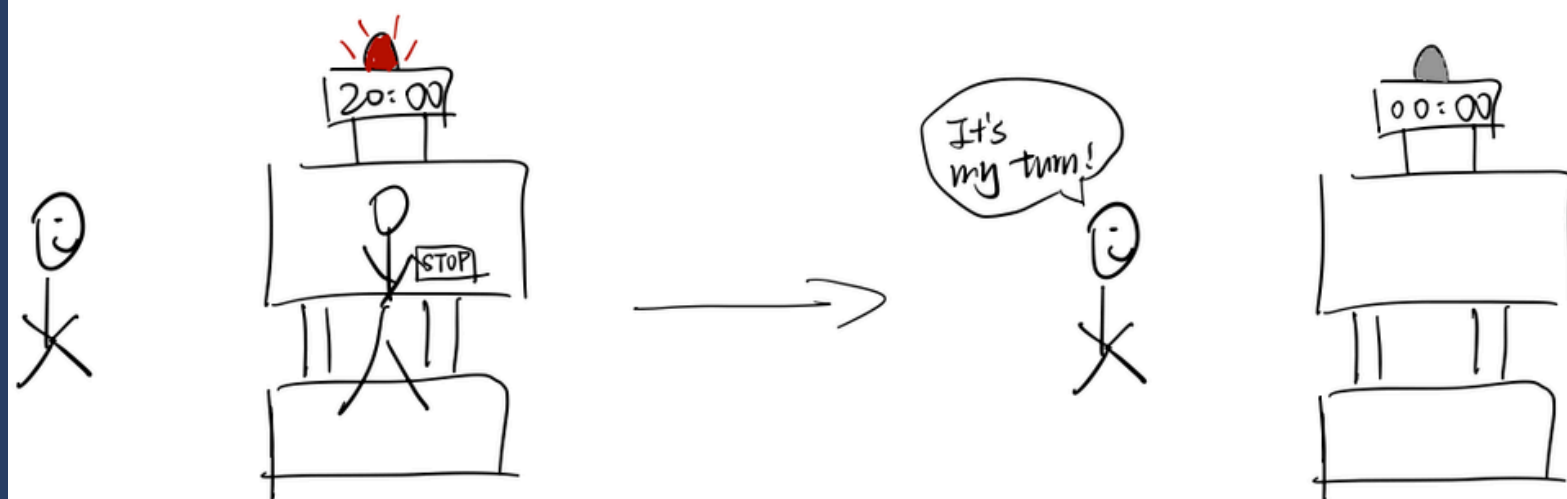
Addresses

- Unclear treadmill usage status
- Inconsistent adherence to the 20-minute policy

Breakdown

- The timer and visual reminders help communicate treadmill usage and support fair equipment sharing..

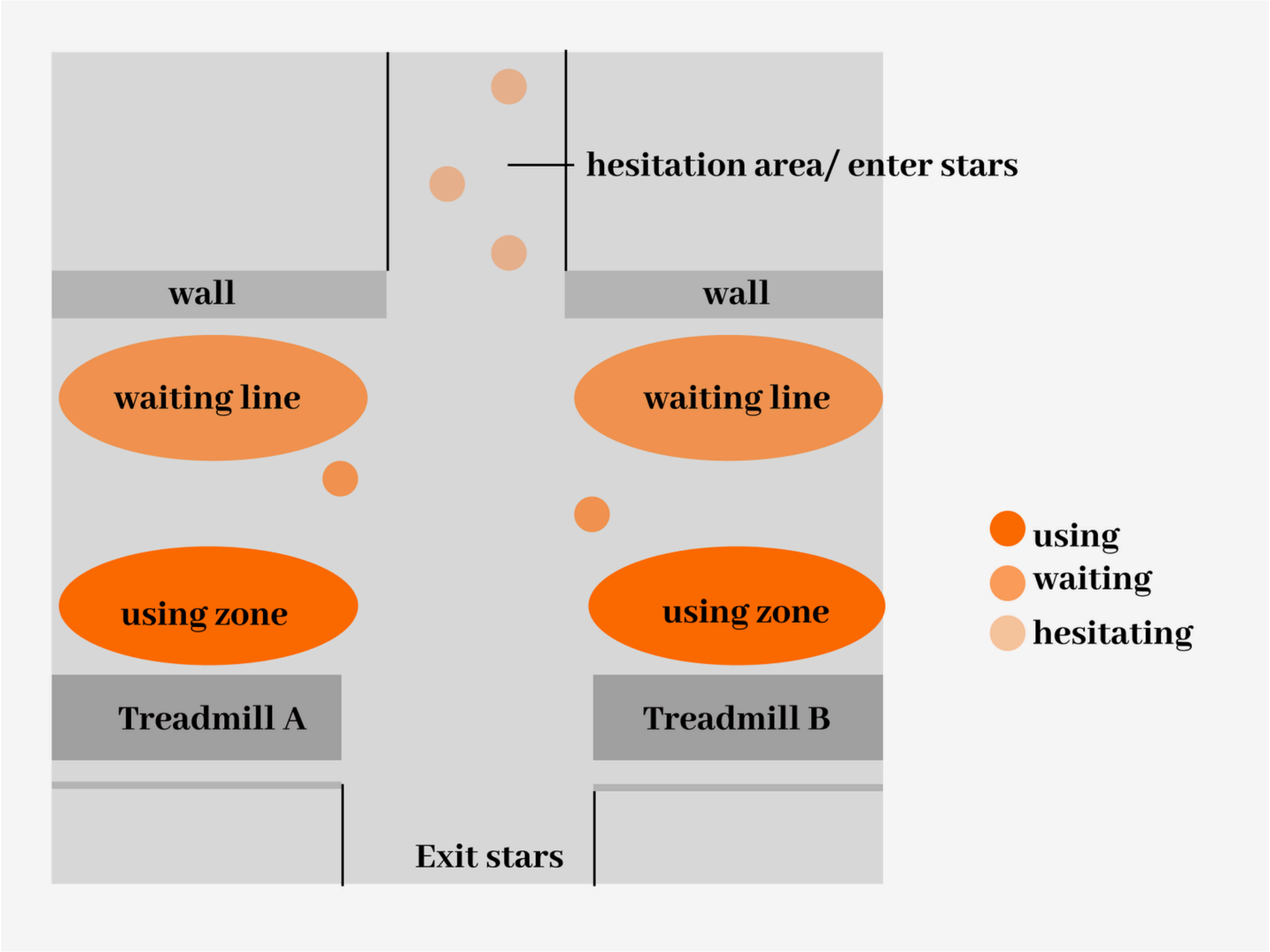
④ Treadmill becomes available.



⑥ Finish and make it available.

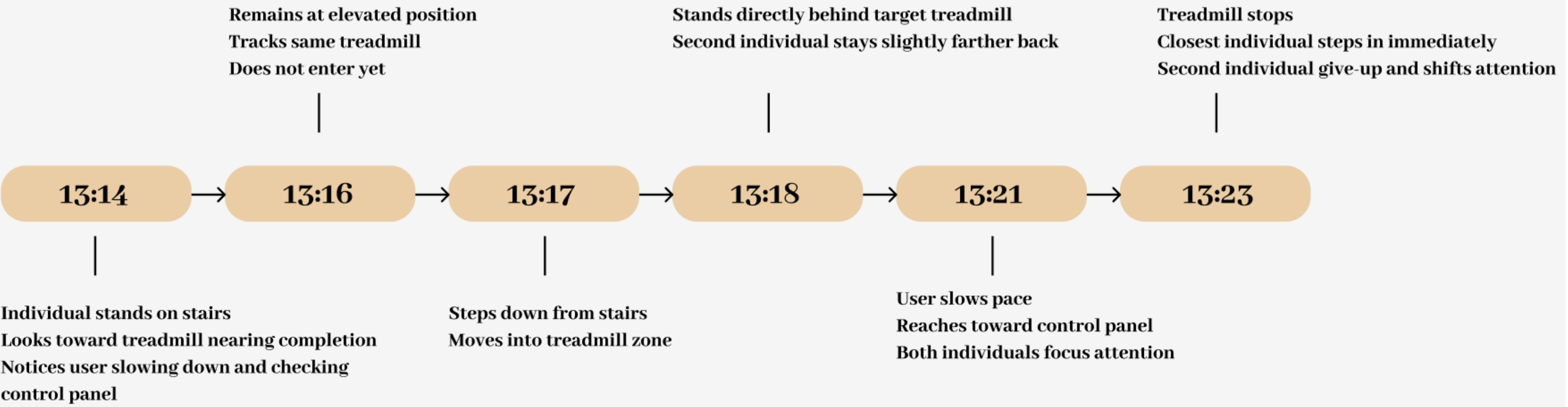


Reference: Activity Maps



Reference: Vertical Timeline

Vertical Timeline



Reference: Affinity Diagram



AI of Byte: Affinity Map

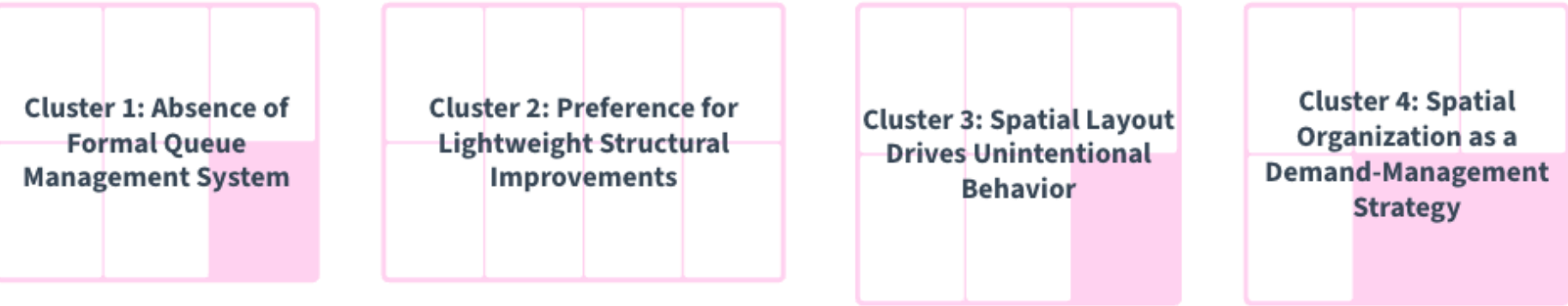
THEME 1: WAITING & AVAILABILITY UNCERTAINTY

People often do not know when a treadmill will open up or how long they will need to wait.



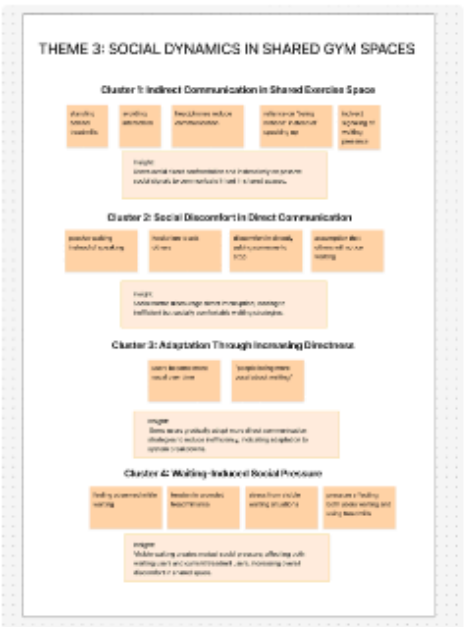
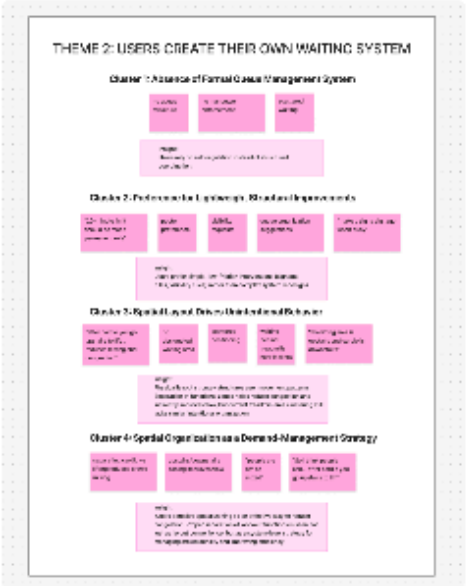
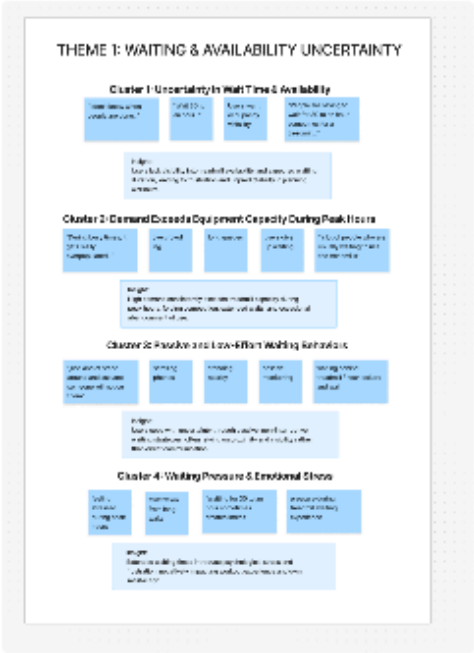
THEME 2: USERS CREATE THEIR OWN WAITING SYSTEM

During busy hours, too many people are trying to use the treadmills at the same time.



THEME 3: SOCIAL DYNAMICS IN SHARED GYM SPACES

Instead of asking directly, people usually wait nearby or keep checking machines from a distance.

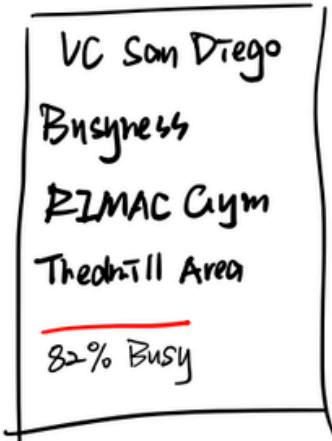


Reference: Story Board

① Check on my phone



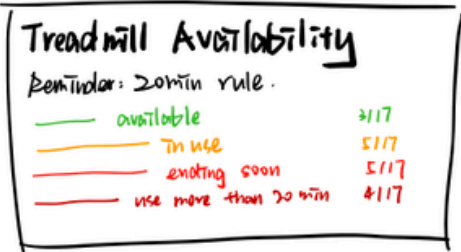
Before I go into RMAC, I go and check the availability of the treadmill area on UCSD mobile app.



② Arrive at RMAC.

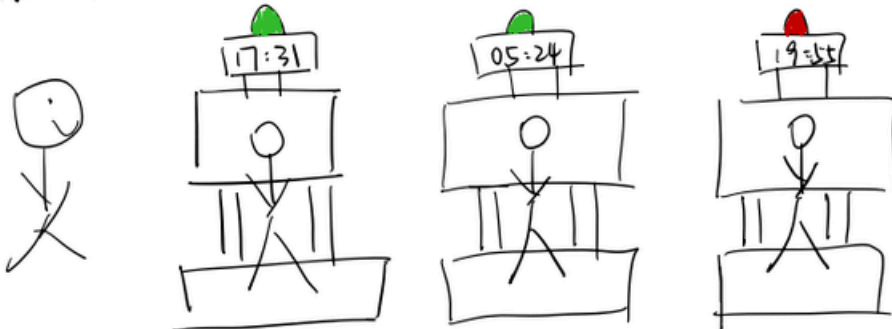


Look at.



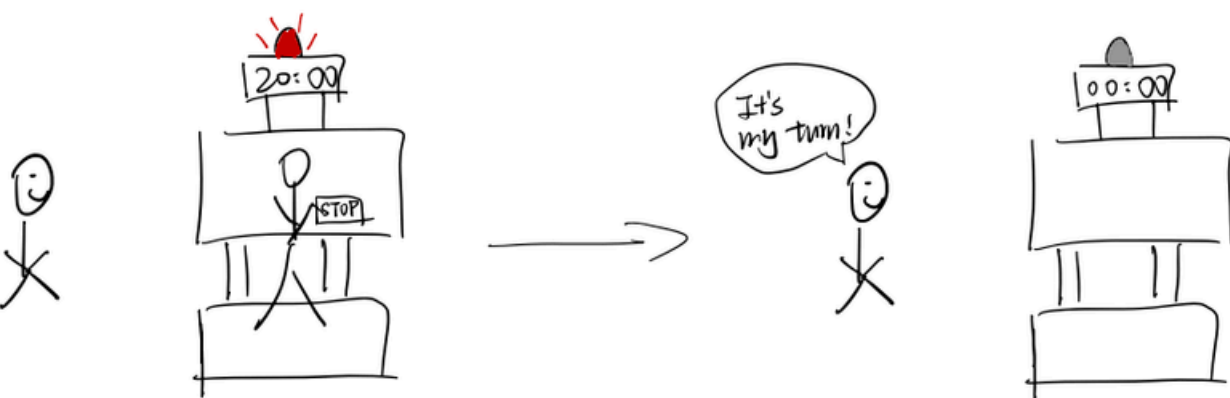
When I arrive at RMAC, I see the system shown on the television of RMAC. I definitely can know the current situation in treadmill zone.

③ Walk into and observe treadmill area.



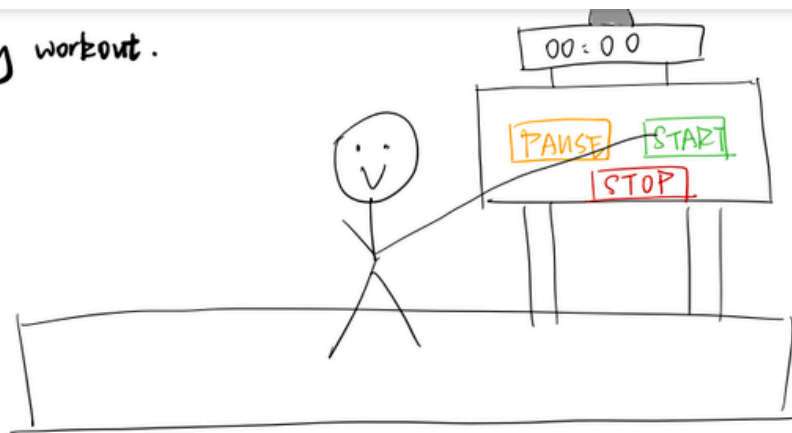
I walk into the treadmill zone, and I want to find an available treadmill as soon as possible. It's easy for me to know how many minutes they have been using. Here, Green Light means users still have time not reaching to 19 mins yet. Red Light means time is almost up. (less than one min reach to 20mins)

④ Treadmill becomes available.



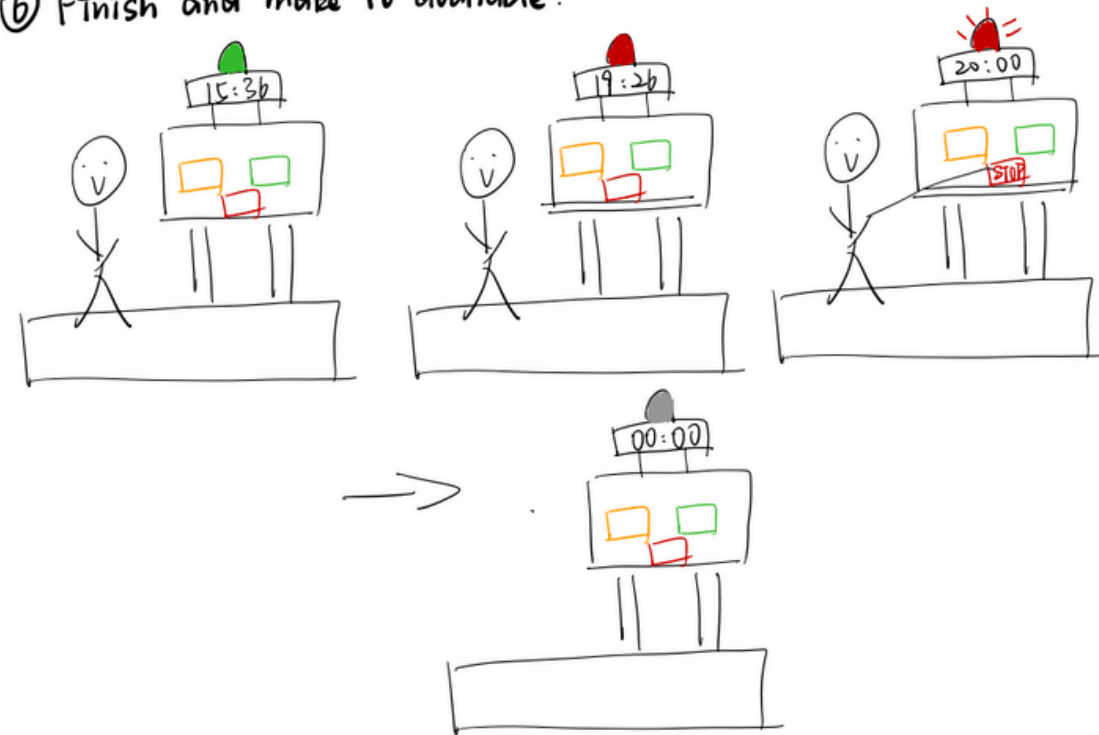
Red Light Flashing means time's up. After the last users press STOP button, the timer will reset, and the light will turn off. And I can enjoy it!

⑤ Start my workout.



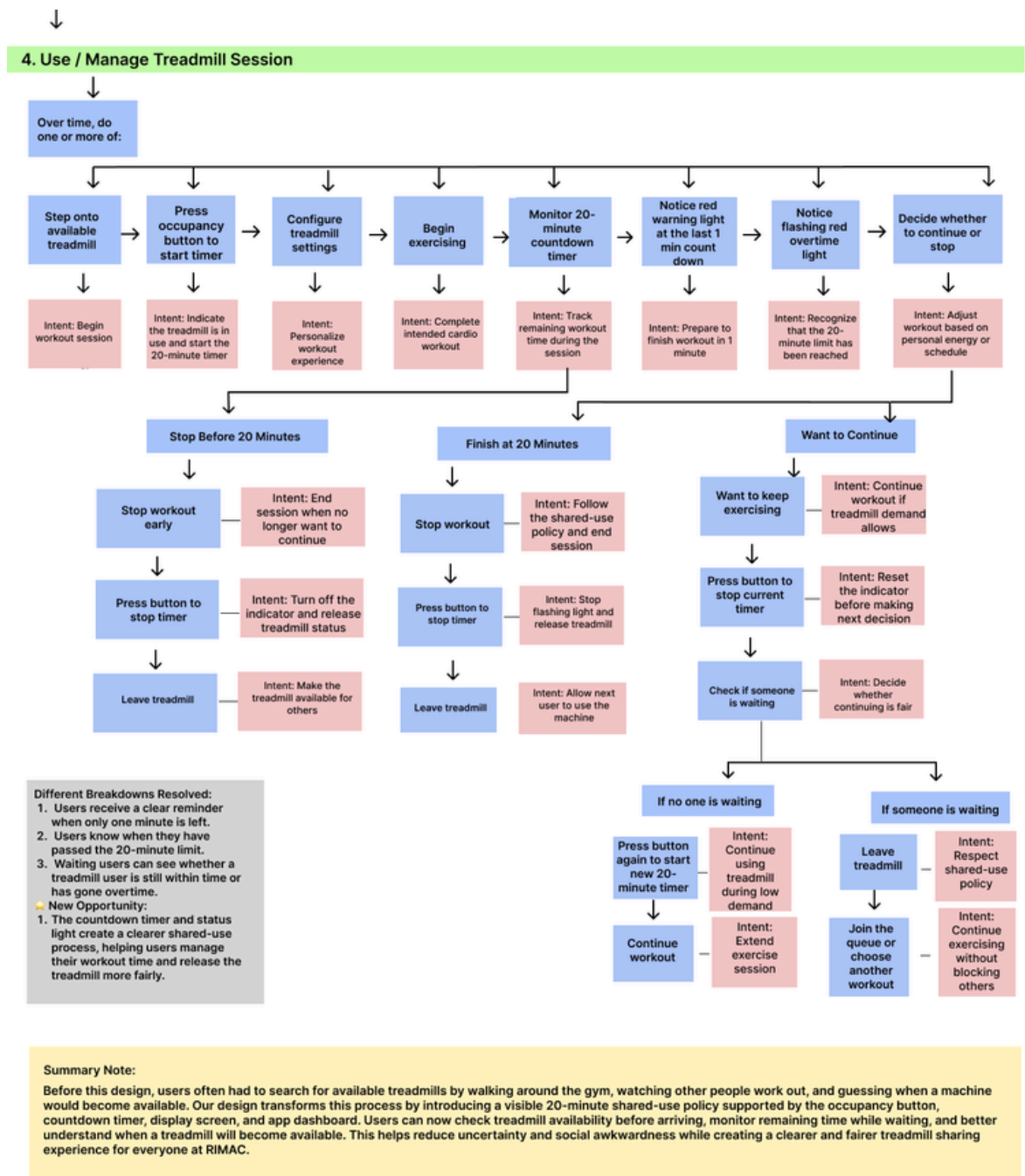
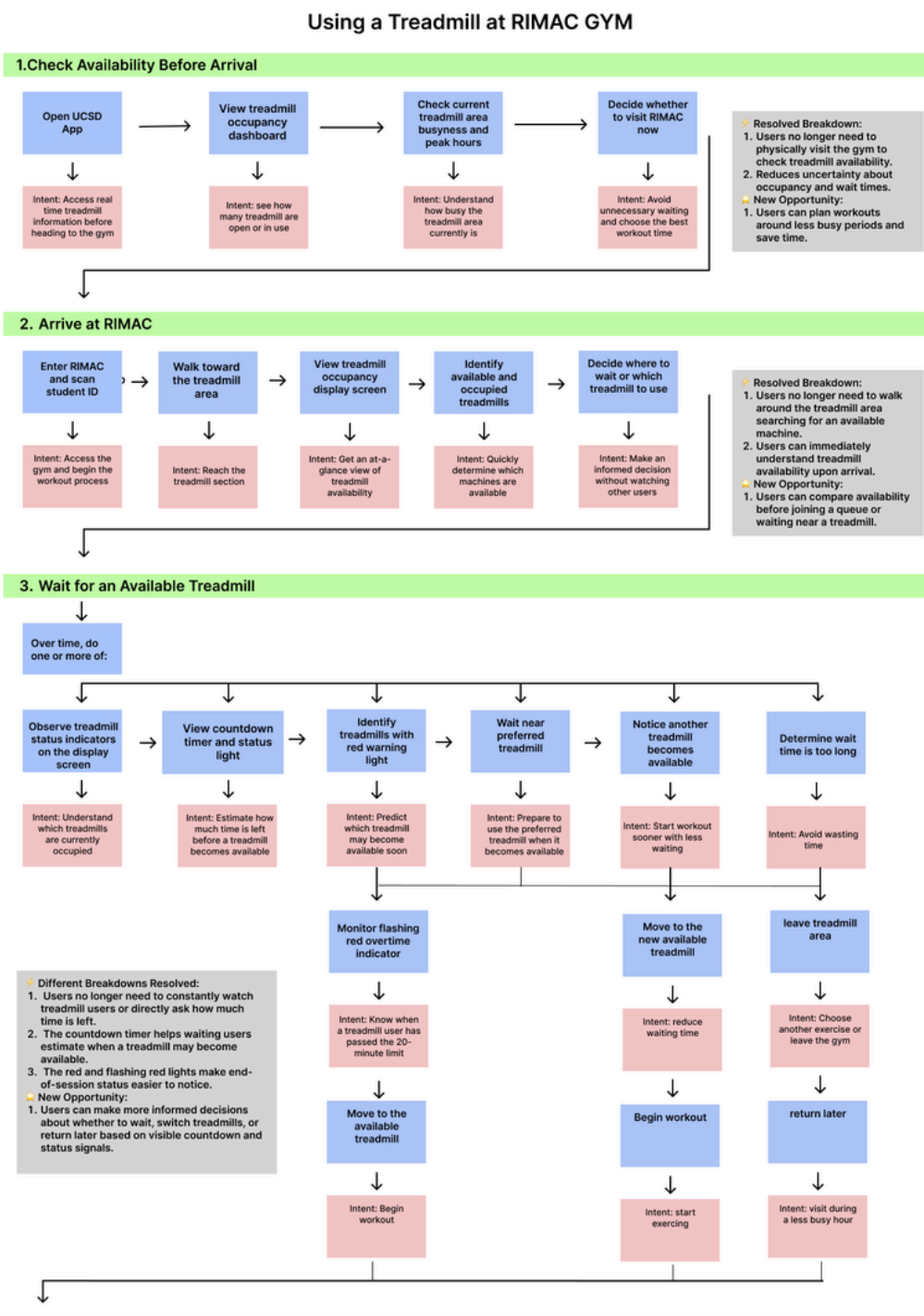
I go and press the button to start my workout and the timer would start counting. The timer and light will still follow the 20-min rule.

⑥ Finish and make it available.



When time reaches to 20 mins, I press STOP, and then leave the treadmill, making it available for other users.

Reference: Sequence modeling



Thank **you**

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